

Geometry & Topology II: Course Notes

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Remarks: These are the notes for Geometry and Topology II, Spring 2026. My personal comments will be in teal.

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1 Vector bundles

1.1 Cotangent bundle

Recall dual vector bundle.

Definition: The cotangent bundle is defined as

$$T^*M \equiv (TM)^*.$$

We call elements of this dual space, *covectors*.

What we are doing here is just taking the dual vector space to T_pM at each point $p \in M$.

Recall: There is a one-to-one correspondence between between tangent vectors at p and derivations at p .

$$\{\text{vectors} \in T_pM\} \longleftrightarrow \{v : C^\infty(M) \rightarrow \mathbb{R} : v(fg)(p) = f(p)v(g) + g(p)v(f)\}$$

Notice that given a point $p \in M$, a covector $\omega \in T_p^*M$, and a tangent vector $v \in T_pM$, then one can define a pairing $\langle \omega, v \rangle \in \mathbb{R}$. This follows basically from the definition of dual vector space. Now, given a function f locally defined near $p \in M$ and a tangent vector $v \in T_pM$, one can similarly define a pairing given by $\langle df_p, v \rangle = v(f)_p \in T_p^*M$, where v is considered as a derivation defined near $p \in M$. We now give the precise definition of the differential.

Definition: Let $f : U \rightarrow \mathbb{R}$ be a smooth function defined on an open neighborhood of $p \in M$. Then the differential (of f at $p \in M$) is defined as

$$\begin{aligned} df_p : T_pM &\rightarrow T_p^*M \\ v &\mapsto df_p(v) \equiv v(f)_p \end{aligned}$$

The differential assigns a scalar to each tangent vector via the pairing in the previous comment.

We can compute the differential in local coordinates as follows:

Given a local chart (U, x) near p , then each coordinate function x^i is smooth and has differential $dx_i(p)$. Moreover, the differentials $\{dx_1(p), \dots, dx_n(p)\} \subset T_p^*M$ provide a dual basis to $\{\frac{\partial}{\partial x^1}(p), \dots, \frac{\partial}{\partial x^n}(p)\} \subset T_pM$. In these local coordinates, we can write the differential as

$$df_p = \sum \frac{\partial f}{\partial x_i}(p) dx_i(p)$$

Lemma: $dx_i(p), \dots, dx_n(p)$ is a basis for T_p^*M .

Remarks:

1. All covectors are differentials
2. For each smooth function $f : M \rightarrow \mathbb{R}$, the map

$$df : M \rightarrow T^*M$$

$$p \mapsto df|_p \in T_p^*M$$

is a smooth section.

Properties: Let $f, g \in C^\infty(M)$. Then

1. $d(af + bg) = a df + b dg$
2. $d(fg) = g df + f dg$
3. $F : N \rightarrow M$ smooth map between smooth manifolds,

$$d(f \circ F)(V) = df(DF(v)) \forall v \in T_q N, q \in N.$$

Lemma: $f \in C^\infty(M)$, $f : M \rightarrow \mathbb{R}$ smooth, $\gamma : [0, 1] \rightarrow M$ smooth path. Then

$$F(\gamma(1)) - F(\gamma(0)) = \int_0^1 df(\gamma'(t)) dt.$$

Definition: A 1-form is a smooth section ω of T^*M . If $F : N \rightarrow M$ is a smooth function, then the pullback of ω is $F^*\omega$ on N , satisfying

$$F^*\omega(v) = \omega(DF(v)), \quad \forall v \in T_p M.$$

Definition: $\gamma : [a, b] \rightarrow M$ be a smooth map. The line integral of ω along γ is

$$\int_a^b \omega(\gamma'(t)) dt.$$

In general: $\omega(p) = \sum \omega(p) \frac{\partial}{\partial x_i} dx_i$

Examples: (1) $M = \mathbb{R}^2$ and consider $\omega = x dy$ which is not exact (not even close). Take the path $\gamma : [0, \pi] \rightarrow \mathbb{R}^2$ given by $\gamma(t) = (\cos(t), \sin(t))$. Then

$$\gamma'(t) = \sin(t) \frac{\partial}{\partial x} + \cos(t) \frac{\partial}{\partial y}$$

$$\omega(\gamma'(t)) = \cos^2(t).$$

$$\int_0^1 \omega(\gamma'(t)) dt = \int_0^1 \cos^2(t) dt > 0$$

Now take the path $\Gamma : [0, \pi] \rightarrow \mathbb{R}^2$ defined by $\Gamma(t) = (\frac{t}{\pi}, 0)$